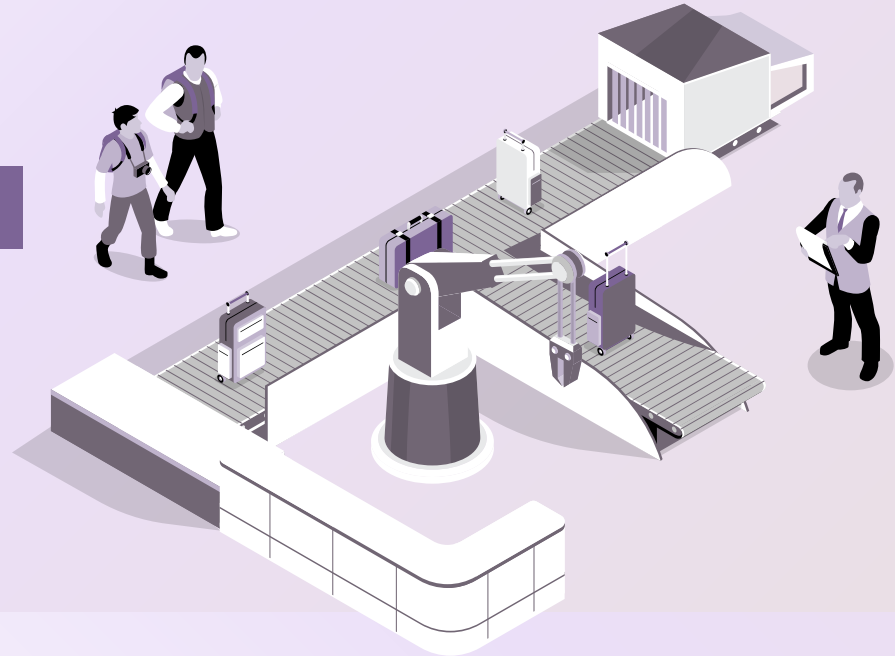


AIRPORT NETWORK

Network Design Case Study

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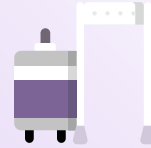


REQUIREMENTS VS DESIGN



REQUIREMENTS

Secure department separation, shared services, Wi-Fi and managed IP addressing.



OUR DESIGN

9 VLANs, router-on-a-stick, DHCP on R1, OSPF to R2-Edge, servers in VLAN 90, WPA2 Wi-Fi.



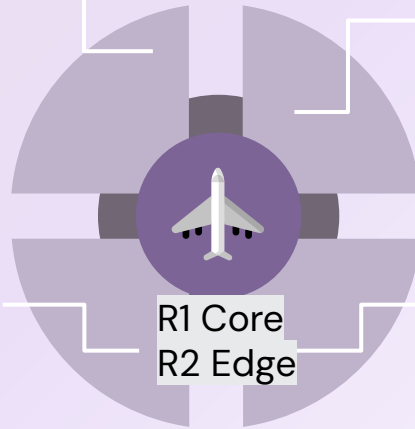
Network Topology by Department

ADMINISTRATION

Admin and check-in PCs connected via SW1-Admin

IT & SERVICES

IT/Ops PCs, DNS, web and airport servers via SW3-Services



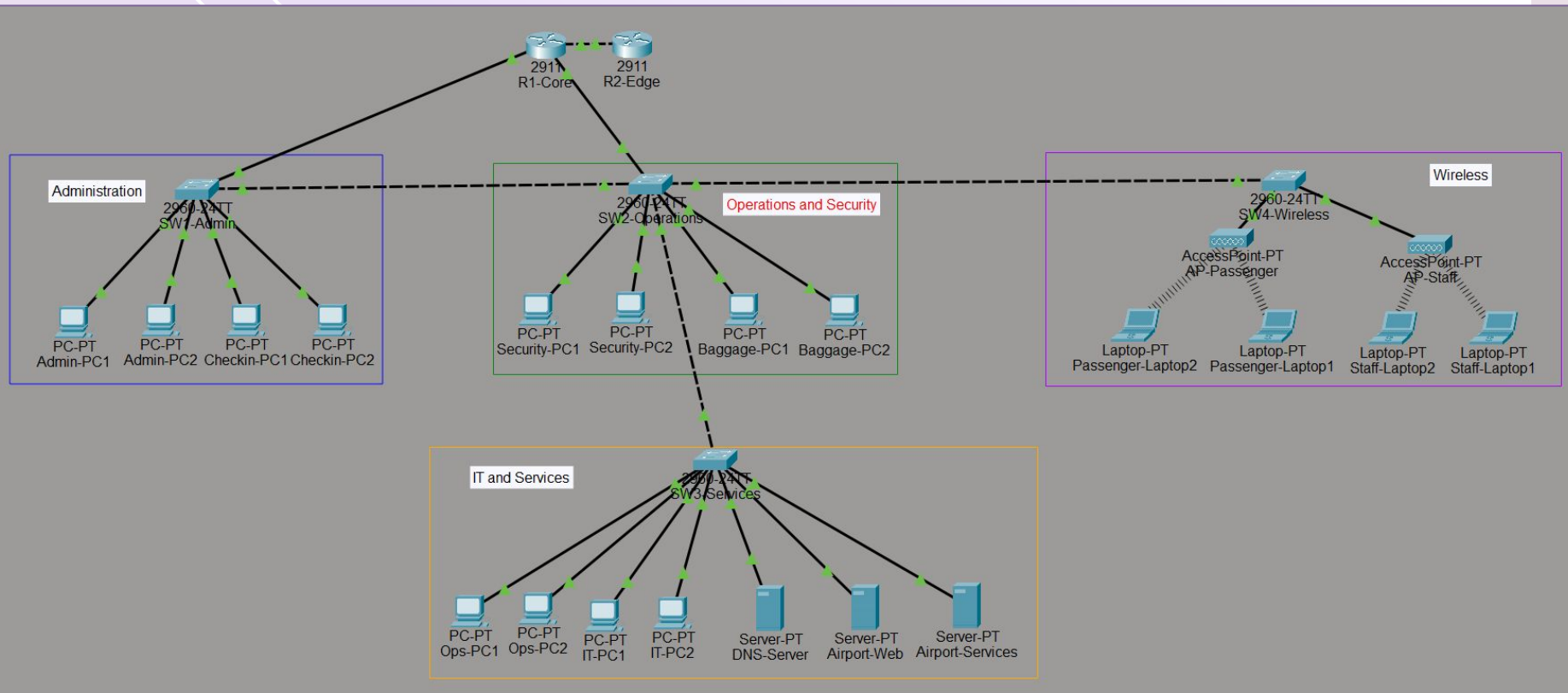
OPERATIONS & SECURITY

Security and baggage PCs connected via SW2-Operations

WIRELESS

Passenger and staff Wi-Fi via SW4-Wireless and 2 APs

NETWORK TOPOLOGY



DESIGN APPROACH



VLANs

9 VLANs: Admin,
Check-in, Security,
Baggage, Ops, IT, Wi-Fi
x2, Servers

INTER-VLAN

R1-Core 802.1Q
subinterfaces
(router-on-a-stick)

ADDRESSING

One /24 per VLAN.
DHCP for clients,
static IPs for servers

ROUTING

OSPF between
R1-Core and R2-Edge
over 10.0.0.0/30

SERVICES

DNS, Web, FTP and
Email hosted in server
VLAN 90

WIRELESS

Staff (VLAN 70) and
Passenger (VLAN 80)
SSIDs on 2 APs

VLAN AND IP ADDRESSING

VLAN	Department	Subnet	Gateway (R1-Core)	Access switch
10	Administration	192.168.10.0/24	192.168.10.1	SW1-Admin
20	Check-in & Ticketing	192.168.20.0/24	192.168.20.1	SW1-Admin
30	Security	192.168.30.0/24	192.168.30.1	SW2-Operations
40	Baggage Handling	192.168.40.0/24	192.168.40.1	SW2-Operations
50	Airport Operations	192.168.50.0/24	192.168.50.1	SW3-Services
60	IT Department	192.168.60.0/24	192.168.60.1	SW3-Services
70	Staff Wi-Fi	192.168.70.0/24	192.168.70.1	SW4-Wireless
80	Passenger Wi-Fi	192.168.80.0/24	192.168.80.1	SW4-Wireless
90	Servers	192.168.90.0/24	192.168.90.1	SW3-Services

- **Router link:** R1 Gi0/0 10.0.0.1/30 – R2 Gi0/0 10.0.0.2/30
- **Servers (static):** DNS 192.168.90.10 · Web 192.168.90.20 · Airport-Services 192.168.90.30
- **DHCP:** first 10 addresses of each client subnet excluded; DNS option 192.168.90.10

SWITCHING: VLANS AND TRUNKS

- **Local VLAN configuration:** each switch defines the VLANs its own hosts need: SW1 (10, 20), SW2 (30–90), SW3 (50, 60, 90), SW4 (70, 80)
- **Local access ports:** hosts attach as untagged access ports, e.g. SW1 Fa0/3–4 in VLAN 10, SW2 Fa0/5–6 in VLAN 30 and Fa0/7–8 in VLAN 40, SW3 Fa0/6–8 in VLAN 90
- **Trunk transport:** SW2 Fa0/1–4 run IEEE 802.1Q and carry tagged frames for the active VLANs between devices
- **Result:** VLAN membership is set per switch, trunks carry it across the network, and R1-Core routes between VLANs

SW2 trunk ports

```
SW2-Operations(config)#do show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     on        802.1q         trunking      1
Fa0/2     on        802.1q         trunking      1
Fa0/3     on        802.1q         trunking      1
Fa0/4     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/1     1-1005
Fa0/2     1-1005
Fa0/3     1-1005
Fa0/4     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,30,40,50,60,70,80,90
Fa0/2     1,30,40,50,60,70,80,90
Fa0/3     1,30,40,50,60,70,80,90
Fa0/4     1,30,40,50,60,70,80,90

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,30,40,50,60,70,80,90
Fa0/2     none
Fa0/3     none
--More--
```

SW3 VLAN assignment

```
SW3-Services(config)#do show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/9, Fa0/10, Fa0/11, Fa0/12,
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16,
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20,
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24,
                                           Gig0/1, Gig0/2
50   OPERATIONS              active    Fa0/2, Fa0/3
60   IT                      active    Fa0/4, Fa0/5
90   SERVERS                 active    Fa0/6, Fa0/7, Fa0/8
1002 fddi-default            active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default         active

SW3-Services(config)#
```

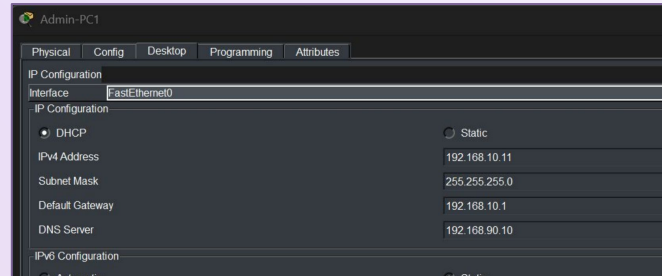
INTER-VLAN ROUTING AND DHCP

R1-Core subinterfaces (gateways)

```
Router(config)#do show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    10.0.0.1        YES manual up          up
GigabitEthernet0/1    unassigned      YES unset  up          up
GigabitEthernet0/1.10 192.168.10.1    YES manual up          up
GigabitEthernet0/1.20 192.168.20.1    YES manual up          up
GigabitEthernet0/2    unassigned      YES unset  up          up
GigabitEthernet0/2.30 192.168.30.1    YES manual up          up
GigabitEthernet0/2.40 192.168.40.1    YES manual up          up
GigabitEthernet0/2.50 192.168.50.1    YES manual up          up
GigabitEthernet0/2.60 192.168.60.1    YES manual up          up
GigabitEthernet0/2.70 192.168.70.1    YES manual up          up
GigabitEthernet0/2.80 192.168.80.1    YES manual up          up
GigabitEthernet0/2.90 192.168.90.1    YES manual up          up
Vlan1             unassigned      YES unset  administratively down down
Router(config)#
```

- **Router-on-a-stick:** Gi0/1.10 and .20 serve VLANs 10 and 20; Gi0/2.30–.90 serve VLANs 30–90. Each subinterface is the .1 gateway of its /24
- **DHCP on R1:** one pool per VLAN with default-router, DNS 192.168.90.10 and the first 10 addresses excluded
- **Outcome:** clients get addresses automatically, e.g. Admin-PC1 192.168.10.11 and Staff-Laptop1 192.168.70.11

Admin-PC1 receives its address by DHCP



ROUTING: OSPF

- **Point-to-point link:** R1-Core 10.0.0.1/30 and R2-Edge 10.0.0.2/30
- **OSPF** advertises all nine VLAN subnets from R1-Core to R2-Edge
- **R2-Edge** learns 192.168.10.0 to 192.168.90.0 /24 as O routes via 10.0.0.1

show ip route on R2-Edge

```
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/30 is directly connected, GigabitEthernet0/0
L    10.0.0.2/32 is directly connected, GigabitEthernet0/0
O    192.168.10.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.20.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.30.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.40.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.50.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.60.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.70.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.80.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.90.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
```

9

VLAN routes learned by R2

100%

R1-R2 ping success (5/5)

/30

10.0.0.0 link subnet

SHARED SERVICES●●●



DNS SERVER

192.168.90.10 · DNS (UDP 53). Resolves airport.local, www.airport.local and services.airport.local



WEB SERVER

192.168.90.20 · HTTP (TCP 80). Airport portal opens at <http://airport.local>



FTP + EMAIL

Airport-Services
192.168.90.30 · FTP (TCP 21) and Email (SMTP/POP3, airport.local)

WIRELESS ACCESS



STAFF WI-FI

SSID Airport-Staff · VLAN 70 ·
WPA2-PSK (AES) · channel 6.
Laptops get 192.168.70.x by
DHCP



PASSENGER WI-FI

SSID Airport-Passenger · VLAN
80 · WPA2-PSK (AES) · channel
6. Gateway 192.168.80.1



CONNECTIVITY TESTING

Source	VLAN	Gateway	DNS server .90.10	Web / Services
Admin-PC1	10	✓	✓	–
Check-in PC	20	✓	✓	–
Security-PC1	30	✓	✓	✓ ping + browser
Baggage-PC1	40	✓	✓	–
Ops-PC1	50	✓	✓	–
IT-PC1	60	✓	✓	–
Staff-Laptop1	70	✓	✓	✓ web .90.20, services .90.30
Passenger-Laptop1	80	✓	✓	✓ browser
Passenger-Laptop2	80	✓	✓	–

- **Ping results:** All recorded ping tests completed successfully (4/4 replies, 0% packet loss).
- **TTL 127** on server pings shows the packets crossed R1-Core (routed between VLANs)

APPLICATION-LAYER SERVICES

DNS (UDP 53)

```
C:\>nslookup airport.local

Server: [192.168.90.10]
Address: 192.168.90.10

Non-authoritative answer:
Name:   airport.local
Address: 192.168.90.20
```

nslookup airport.local returns 192.168.90.20

HTTP (TCP 80)



Portal opens by name from staff, security and passenger devices

FTP (TCP 21)

```
C:\>
C:\>ftp 192.168.90.30
Trying to connect...192.168.90.30
Connected to 192.168.90.30
220- Welcome to PT Ftp server
Username:airportadmin
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>
```

Login to 192.168.90.30 succeeds (230 Logged in)

- **Email:** SMTP and POP3 enabled on Airport-Services, domain airport.local, users admin and ops

SIMULATION MODE: PACKET TRACE

DNS query (UDP): Staff-Laptop 1 to DNS-Server

PDU Information at Device: DNS-Server

OSI Model | Inbound PDU Details | Outbound PDU Details

At Device: DNS-Server
Source: Staff-Laptop1
Destination: 192.168.90.10

In Layers	Out Layers
Layer 7: DNS	Layer 7: DNS
Layer 6	Layer 6
Layer 5	Layer 5
Layer 4: UDP Src Port: 1027, Dst Port: 53	Layer 4: UDP Src Port: 53, Dst Port: 1027
Layer 3: IP Header Src. IP: 192.168.70.11, Dest. IP: 192.168.90.10	Layer 3: IP Header Src. IP: 192.168.90.10, Dest. IP: 192.168.70.11
Layer 2: Ethernet II Header 00E0.8FDA.C803 >> 00D0.BA19.701C	Layer 2: Ethernet II Header 00D0.BA19.701C >> 00E0.8FDA.C803
Layer 1: Port FastEthernet0	Layer 1: Port(s): FastEthernet0

- **App/Transport:** DNS over UDP 1027 to 53
- **Network/Link:** 192.168.70.11 to 192.168.90.10, Ethernet II frame
- **Path:** AP-Staff, SW4, SW2, R1-Core, SW2, SW3, DNS-Server

HTTP (TCP): Security-PC1 to Airport-Web

PDU Information at Device: Security-PC1

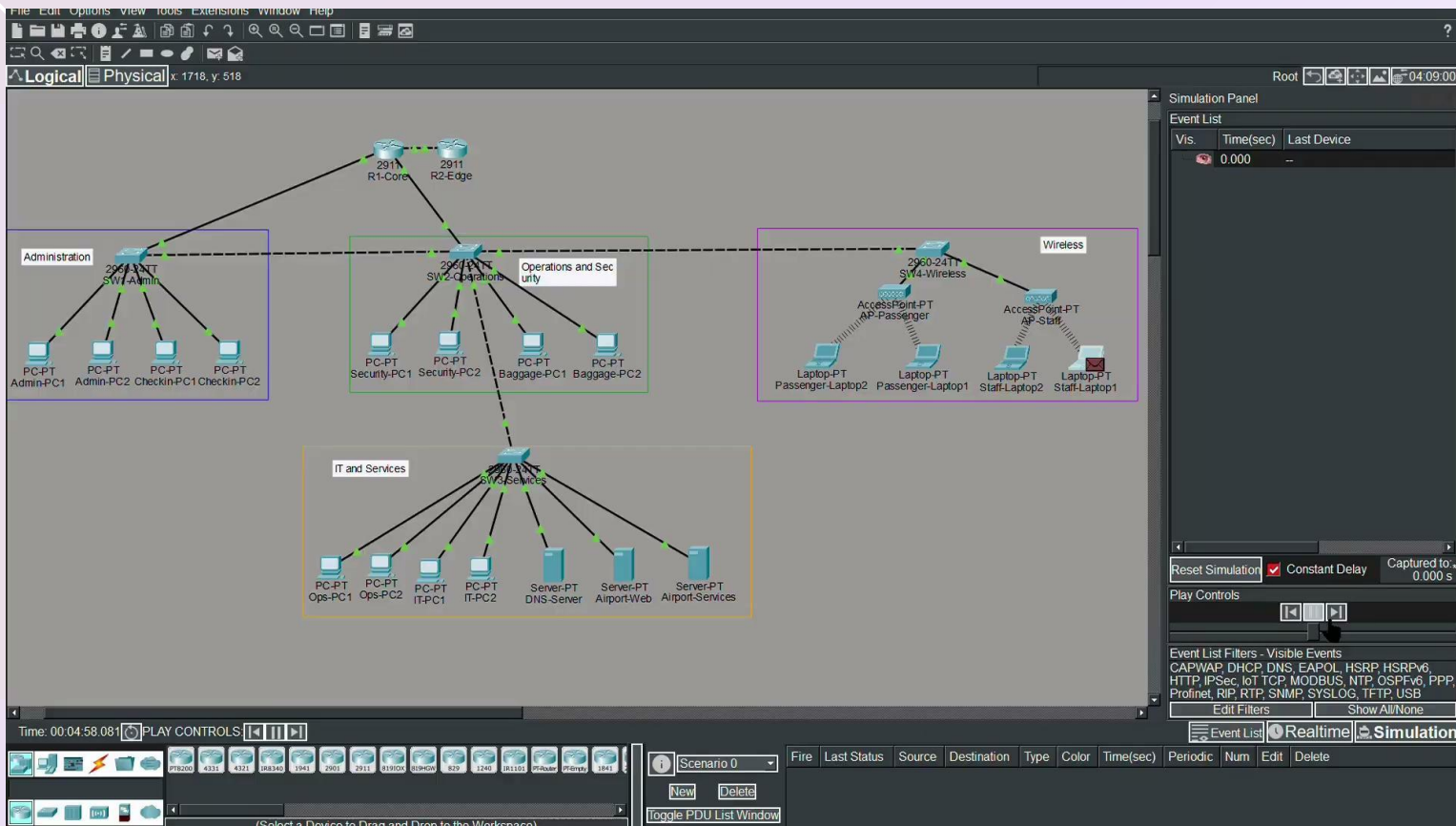
OSI Model | Outbound PDU Details

At Device: Security-PC1
Source: Security-PC1
Destination: 192.168.90.20

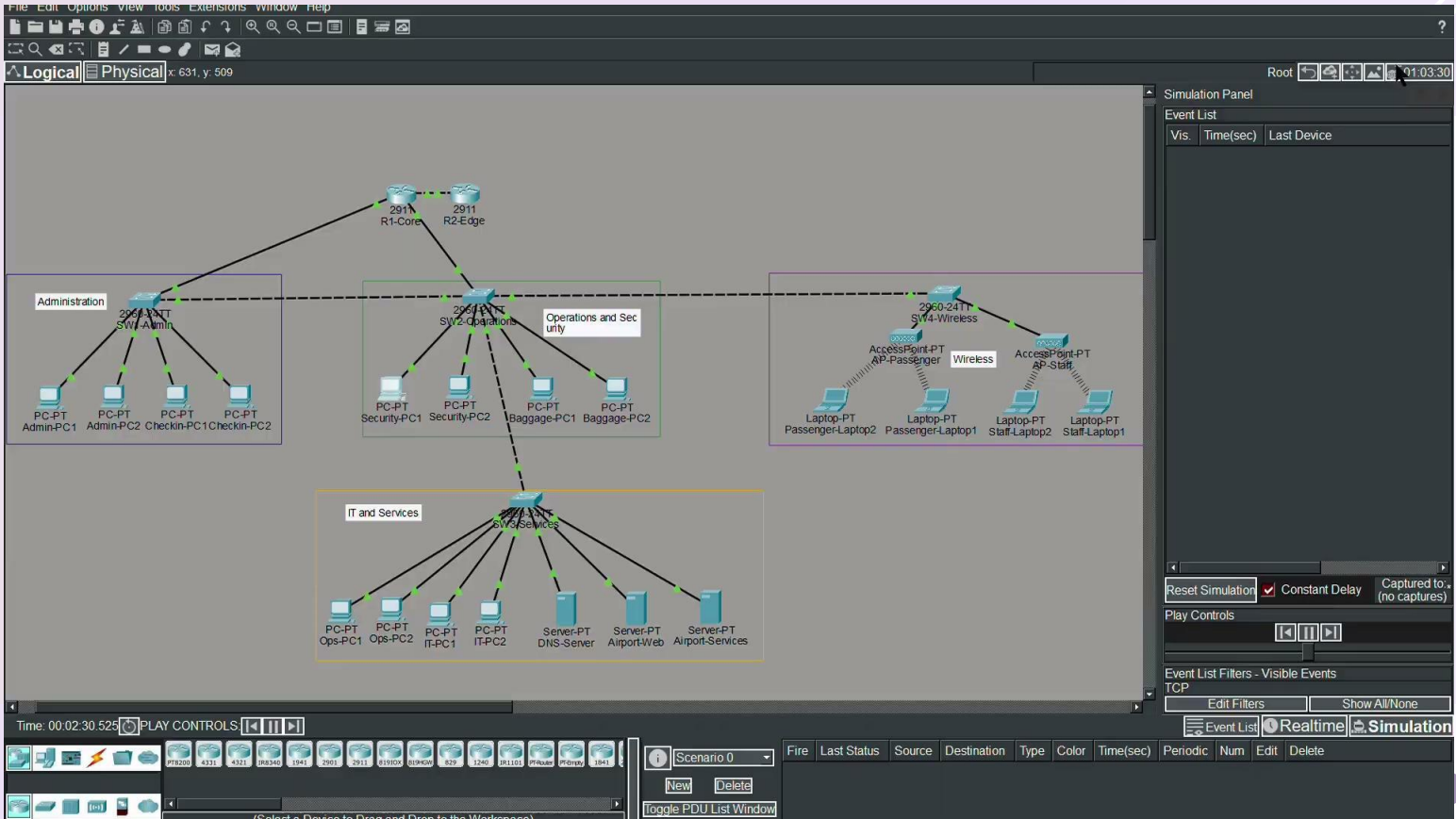
In Layers	Out Layers
Layer 7	Layer 7
Layer 6	Layer 6
Layer 5	Layer 5
Layer 4	Layer 4: TCP Src Port: 1026, Dst Port: 80
Layer 3	Layer 3: IP Header Src. IP: 192.168.30.11, Dest. IP: 192.168.90.20
Layer 2	Layer 2: Ethernet II Header 0002.4A71.EA27 >> 00E0.8FDA.C803
Layer 1	Layer 1: Port(s):

- **App/Transport:** HTTP over TCP 1026 to 80, connection opened then closed (FIN+ACK)
- **Network/Link:** 192.168.30.11 to 192.168.90.20, Ethernet II frame
- **Routing:** R1-Core forwards between VLAN 30 and VLAN 90

DNS Packet Trace Simulation



HTTP TCP Packet Trace Simulation



THANK YOU

